

3. Functional Requirements

3.1 Actor Profiles

An actor will be an external entity that interacts with HANSRAJ QUERY BRIDGE to perform a specific function or to exchange information with the system. Actors may be human users or external software services.

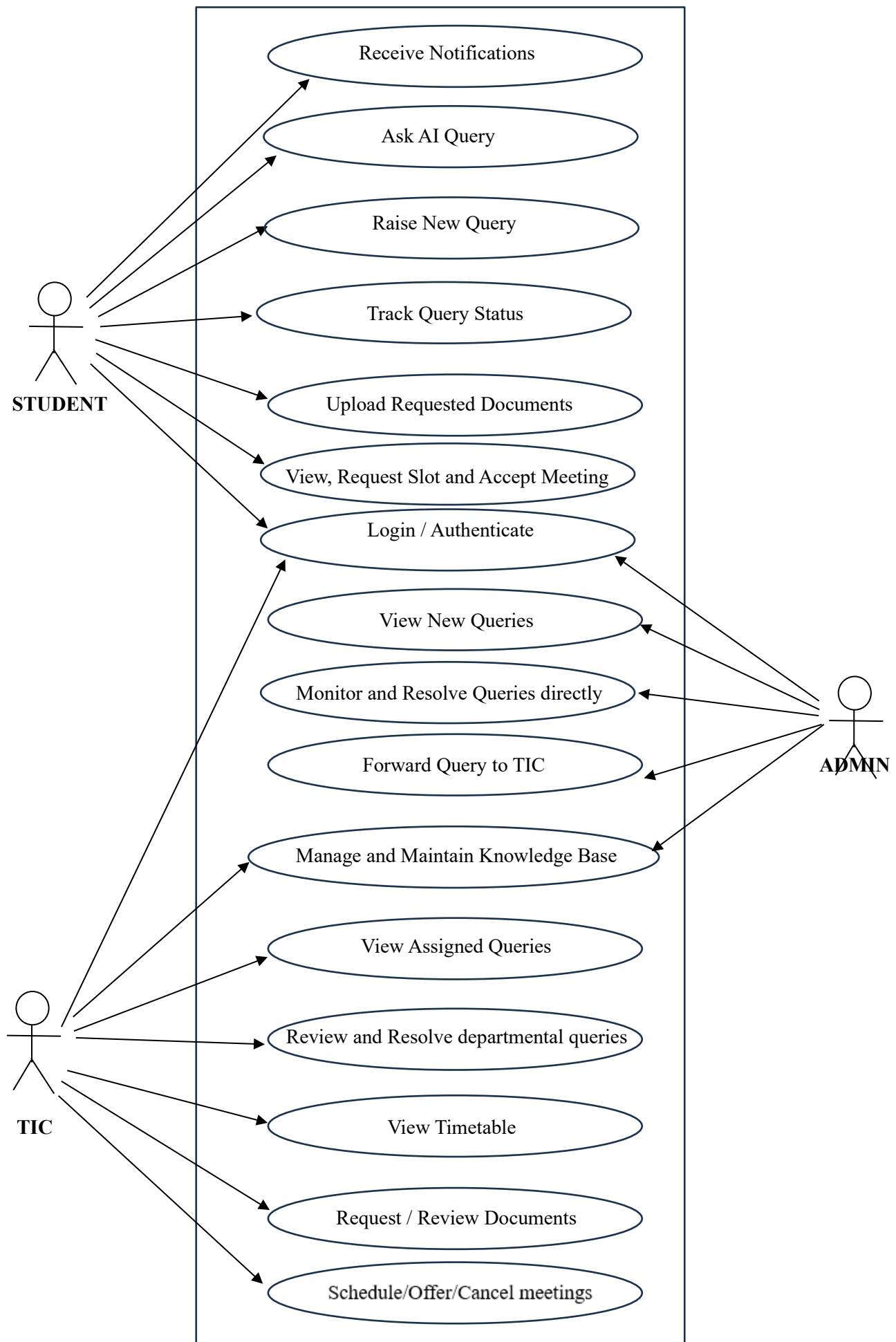
The actors will be broadly classified into:

- **Primary Actors:** Directly interact with HANS Connect to accomplish their tasks.
- **Supporting Actors:** Provide supporting functionality required by the system.

The following actors are identified for the system:

Actor	Type	Description
Student	Primary	The primary end user who uses this system to ask questions, raise unresolved queries, track query status, receive resolutions, upload requested documents, and schedule or respond to meetings with TICs.
Admin	Primary	The authorized college administrative user responsible for reviewing incoming queries, resolving administrative queries, forwarding department-specific queries to TICs, and managing verified institutional knowledge.
TIC	Primary	The faculty member responsible for handling queries related to a particular department. The TIC can resolve assigned queries, request and review documents, view timetable availability, schedule student meetings, and update department-specific knowledge.
AI / RAG Engine	Supporting	The AI component responsible for understanding student queries, performing semantic retrieval from the verified knowledge base, generating context-aware responses, and determining when a query should be escalated due to insufficient or low-confidence information.
Authentication Service	Supporting	An external college authentication service responsible for validating user credentials and providing authenticated user information and role details.
Notification Service	Supporting	An external email service responsible for delivering notifications regarding query updates, resolutions, document requests, document submissions, and meeting schedules.
Document Storage Service	Supporting	The storage service responsible for securely storing and retrieving student-uploaded documents and associated metadata.
Timetable Service	Supporting	A service or institutional data source used to provide TIC timetable information and available slots for scheduling student meetings.

3.2 Use – Case Diagram



3.3 Student Use Cases

1. Use Case: Login / Authenticate Schedule/Offer/Cancel meetings

I. Introduction: The Authenticate/Login use case allows registered users to securely access HANSRAJ QUERY BRIDGE using their authorized college credentials. The system verifies the user's identity and role and provides access to the appropriate dashboard.

II. Actors:

1. **Primary Actors:** Students
2. **Supporting Actors:** College authentication service

III. Pre - Conditions:

1. The user must have valid college login credentials.
2. The user must have an active account and role in the college official database.
3. HANSRAJ QUERY BRIDGE system and authentication services must be available.

IV. Post-Conditions:

1. The user's identity is successfully verified.
2. The user is redirected to the dashboard according to their assigned role.
3. A secure authenticated session is established.

V. Flow of Events:

Basic Flow

1. The user opens the HANSRAJ QUERY BRIDGE login page.
2. The system displays the login interface.
3. The user enters their official college credentials.
4. The system sends the login credentials to the authentication service for verification.
5. The authentication service validates the credentials.
6. The system retrieves the user's identity and assigned role.
7. The system creates a secure user session.
8. The system applies role-based access permissions.
9. The user is redirected to the appropriate dashboard.

Alternate Flows

A1. Invalid Credentials

1. The authentication service rejects the user-provided credentials.
2. The system displays an appropriate authentication error message.
3. The user is allowed to retry with valid credentials.

A2. Authentication Service Unavailable

1. The system cannot communicate with the authentication service.
2. The system displays a service-unavailable message.
3. No user session is created.

A3. Valid Credentials but Unauthorized Account

1. The credentials are successfully verified.
2. The system finds that the account is inactive or does not have an authorized role.
3. Access to HANSRAJ QUERY BRIDGE is denied.

VI. Special Requirements:

1. Authentication credentials will be transmitted using HTTPS.
2. The system shall implement Role-Based Access Control (RBAC).
3. Authentication sessions shall have appropriate expiration time and security controls.
4. The system shall not reveal sensitive information about why a credential validation failed.

2. Use Case: Ask AI Query

I. Introduction: The Ask AI Query use case allows an authenticated student to submit a natural-language query through the HASRAJ QUERY BRIDGE chatbot. The system analyses the query, retrieves relevant verified information from the institutional knowledge base using RAG and semantic search, and provides a response when a sufficiently relevant and reliable answer is available.

II. Actors:

1. **Primary Actors:** Student
2. **Supporting Actors:** AI/RAG Engine, Knowledge Base

III. Pre - Conditions:

1. The student is successfully authenticated.
2. The AI/RAG service and knowledge base are available.

IV. Post-Conditions:

1. The student's query is processed and recorded.
2. If a relevant verified answer is found, the response is displayed to the student.
3. If no reliable answer is found, the system offers the student an option to raise the query to the administration.
4. The query is stored for system monitoring and future knowledge-base improvement.

V. Flow of Events:**Basic Flow**

1. The student opens the HANSRAJ QUERY BRIDGE chatbot.
2. The student enters a query in natural language.
3. The system validates and preprocesses the submitted query.
4. The AI/RAG Engine analyses the query to identify its intent and relevant context.
5. The system generates an embedding representation of the query.
6. The system performs semantic search against the verified institutional knowledge base.
7. The system retrieves the most relevant knowledge entries.
8. The system computes the relevance and confidence of the retrieved information.
9. If a sufficiently relevant and verified answer is found, the retrieved information is provided to the AI model.
10. The AI generates a response based only on the retrieved institutional information.
11. The response is displayed to the student along with an option to raise a query if further assistance is required.
12. The query and generated response are recorded for tracking and system monitoring.

Alternate Flows**A1. No Sufficiently Relevant Answer Found**

1. The semantic search does not find information meeting the required relevance threshold.
2. The system does not generate an unsupported answer.
3. The system informs the student that a verified answer is currently unavailable.
4. The system displays an option to Raise Query to Admin.
5. The student may proceed to **Use Case 3: Raise New Query**.

A2. Incomplete Query

1. The system detects that the query does not contain sufficient information to determine the student's intent.
2. The system asks follow-up questions to the student for clarification.
3. The student provides additional information.
4. The system processes the updated query again.

A3. Malicious or Invalid Input

1. The system detects potentially malicious, abusive, or invalid input.
2. The system rejects the input.
3. The student is informed that the submitted query cannot be processed.
4. Repeated suspicious activity may be logged.

A4. AI Response Generation Failure

1. The required information is successfully retrieved.
2. The AI model fails to generate a response.
3. The system informs the student that the response could not be generated.
4. The student may retry or raise the query to the administration.

A5. Student Requests Further Clarification

1. The student indicates that the response does not fully address their question.
2. The student submits a follow-up query.
3. The system processes the follow-up query using the existing conversation.
4. A new response is generated if sufficient verified information is available.

VI. Special Requirements :

1. The system shall use verified institutional information as the primary source for AI-generated responses.
2. The AI shall not provide an answer when the retrieved information does not meet the defined relevance threshold.
3. The system shall use semantic similarity rather than relying only on exact keyword matching.
4. The system should support natural-language queries, including variations in wording and commonly used terms.
5. The system shall implement input validation, rate limiting, and safeguards against malicious prompts attempts.
6. Sensitive student information shall not be unnecessarily exposed to the AI model or other users.

3. Use Case: Raise New Query

I. Introduction: Raise New Query use case allows a student to submit a query to the Administrator when the AI system cannot provide a sufficiently reliable or relevant answer. The submitted query is recorded and made available to the Administrator for review and resolution.

II. Actors:

1. **Primary Actors:** Students
2. **Supporting Actors :** AI/RAG Engine, Admin

III. Pre - Conditions :

1. The student is authenticated.
2. The student has submitted a query through the chatbot.
3. The student has a query that requires human assistance because either the AI could not find sufficiently relevant information or the student chooses to seek administrative assistance.

IV. Post - Conditions :

1. The query is successfully recorded in the system.
2. The query is assigned the appropriate initial status, such as New or Pending Review.
3. The Administrator can view the submitted query.
4. The student can track the query from their dashboard.

V. Flow of Events

Basic Flow

1. Student selects "Raise Query to Admin".
2. System displays the query submission form.
3. Student enters the query and provides additional description or relevant details.
4. System validates the submitted information.
5. Student submits the query.
6. System creates a new query record.
7. System assigns the initial status as New.
8. System makes the query available in the Admin dashboard.
9. The query becomes available for further processing by the Admin.

Alternate Flow

A1. Incomplete or invalid submission

1. Student submits incomplete or invalid information.
2. System displays an appropriate validation message.
3. Student corrects the information and resubmits.

A2. Submission failure

1. System is unable to save the query due to a temporary failure.
2. System informs the student that the submission was unsuccessful.
3. Student may retry the submission.

A3. Student cancels submission

4. Student chooses to cancel before submitting.

1. System discards the unsaved query.
2. Student returns to the previous page.

VI. Special Requirements :

1. The system shall prevent duplicate submissions where appropriate.
2. Student information associated with the query shall only be accessible to authorized users.
3. The submitted query shall be stored securely for tracking and audit purposes.
4. The system should notify the Administrator of newly submitted queries.
5. The system shall provide clear confirmation after successful submission.

4. Use Case : Track Query Status

I. Introduction : Track Query Status use case allows an authenticated student to view the current status and progress of queries submitted to the Admin or TIC. The system provides the student with relevant status information and updates as the query progresses toward resolution.

II. Actors

1. **Primary Actors :** Students
2. **Supporting Actors :** Admin, TIC

III. Pre - Conditions :

1. Student is authenticated.
2. Student has at least one submitted query.

IV. Post - Conditions :

1. The student can view the current status of their submitted query.
2. The student can view relevant progress information, such as whether the query is being reviewed, forwarded, or resolved.
3. The system does not modify the query merely by viewing its status.

V. Flow of Events

Basic Flow

1. Student opens the “Query Status” section.
2. System retrieves the queries submitted by the student.
3. System displays the queries along with their current status.
4. Student selects a particular query.
5. System displays the query details, including submitted query, submission date, current status, assigned authority where applicable and resolution if available.
6. Student can return to the query list or view another query.

Alternate Flow

A1. Query has been forwarded to TIC

1. System shows that the query has been forwarded to the relevant TIC.
2. Student can view the updated status and assigned authority.
3. No further action is required from the student unless the TIC requests additional information or documents.

A2. Query resolution is not yet available

1. Student selects a query that has not been resolved.
2. System informs the student that the resolution is still pending.
3. Student is returned to the query status view.

A3. Query has been resolved

1. System displays the **Resolved** status.
2. Student can select the query to view the resolution.
3. If applicable, the student can view any additional instructions or requested actions.

A4. Status information temporarily unavailable

1. System is unable to retrieve query information.
2. System displays an appropriate error message.
3. Student may retry the request.

VI. Special Requirements

1. Students shall only be able to view queries submitted by themselves.
2. Query status shall reflect the latest update recorded by the Administrator or TIC.
3. Status changes shall be recorded with appropriate timestamps.
4. The system shall provide clear status labels.
5. Sensitive information belonging to other students shall not be exposed.
6. Resolution information shall not be editable by the student.
7. If the resolution contains a request for documents or a meeting, the system shall provide the appropriate next action.

5. Use Case : Upload Requested Documents

I. Introduction : Upload Requested Documents use case allows an authenticated student to securely upload documents requested by the Admin or TIC as part of the query resolution process. The system validates and associates the uploaded documents with the relevant query and makes them available to the authorized authority for review.

II. Actors :

1. **Primary Actors :** Students
2. **Supporting Actors :** TIC/Admins
3. **Supporting Service :** Document Storage Service

III. Pre - Conditions :

1. Student is authenticated.
2. The student has a query requiring document submission.
3. The Admin or TIC has requested one or more documents.
4. The document submission option is enabled for the relevant query.

IV. Post - Conditions :

1. Required documents are successfully uploaded and securely stored.
2. The query status is updated to indicate that the requested documents have been submitted.
3. The concerned Admin or TIC can access the submitted documents.

V. Flow of Events

Basic Flow

1. Student opens the relevant query from My Queries.
2. System displays the document request and the required documents.
3. Student selects the appropriate documents for upload.
4. System validates the selected files.
5. Student submits the documents.
6. System securely uploads and stores the documents.
7. System associates the documents with the corresponding query.
8. System updates the query status to Documents Submitted.
9. System notifies the assigned Administrator or TIC that the documents have been submitted.
10. System displays a confirmation to the student.

Alternate Flow

A1. Invalid file

1. Student selects a file that does not meet the allowed file type, size, or format requirements.
2. System rejects the file.
3. System displays the reason for rejection.
4. Student selects a valid file and retries the upload.

A2. Required document missing

1. Student attempts to submit the document request without providing all mandatory documents.
2. System identifies the missing documents.
3. System asks the student to provide the missing documents.
4. Student uploads the required documents and resubmits.

A3. Upload failure

1. An error occurs while uploading or storing the document. System informs the student that the upload was unsuccessful.
2. Previously submitted documents remain unaffected.
3. Student may retry the upload.

VI. Special Requirements

1. Only authenticated students shall be permitted to upload documents for their own queries.
2. Uploaded files shall be securely stored and protected from unauthorized access.
3. The system shall validate file type, size, and other applicable restrictions before storage.
4. Students shall not be able to access documents submitted by other students.
5. The system shall maintain a link between each document, the student, and the relevant query.
6. The system should clearly indicate which documents are required, submitted, under review, or accepted/rejected.

6. Use Case : View, Request Slot and Accept Meeting

I. Introduction: View, Request Slot and Accept Meeting use case allows a student to view available meeting slots provided by the TIC, request to schedule a suitable meeting, and accept or confirm the scheduled meeting. The system records the meeting details and provides the student and TIC with the relevant meeting information.

II. Actors:

1. **Primary Actors** : Student
2. **Supporting Actors** : TIC
3. **Supporting Service** : Timetable Service, Notification Service

III. Pre - Conditions:

1. Student is authenticated.
2. A query or interaction with the TIC requires a meeting.
3. The TIC has made available one or more meeting slots.
4. Timetable information is available to the system.

IV. Post - Conditions:

1. A meeting is successfully scheduled and associated with the student and relevant query.
2. Student and TIC can view the meeting details.
3. Appropriate notifications are generated.

V. Flow of Events:**Basic Flow**

1. Student opens the “Meetings” section or selects the meeting option from the relevant query.
2. System retrieves the available meeting slots provided by the TIC.
3. System displays the available dates and times.
4. Student selects a suitable meeting slot.
5. System verifies that the selected slot is still available.
6. Student confirms the selected slot.
7. System creates the meeting record.
8. System marks the selected slot as **Booked**.
9. System displays the meeting details to the student.
10. System makes the scheduled meeting visible to the TIC.
11. System sends a notification to the student and TIC.

Alternate Flow**A1. Student rejects or declines a proposed meeting**

1. Student views the proposed meeting.
2. Student selects **Decline/Cancel Meeting**.
3. System records the response.
4. The slot is made available again, where applicable.
5. TIC is notified of the student's response.

A2. Selected slot is no longer available

1. Student selects a meeting slot.
2. Another booking has already occupied the slot.
3. System informs the student that the slot is no longer available.
4. Student selects another available slot.

A3. No meeting slots available

1. System finds no available slots.
2. System informs the student that no meeting slots are currently available.
3. Student may check again later for availability.

VI. Special Requirements

1. Only authenticated students shall be permitted to schedule or respond to meetings.
2. Students shall only see meeting slots relevant to their assigned TIC/query.
3. The system shall display meeting date, time, TIC, and relevant query/reference information.
4. Meeting status shall be clearly indicated, such as **Proposed, Accepted, Booked, Declined, Cancelled, or Completed.**
5. Changes to meeting schedules shall be reflected for both the student and TIC.
6. Meeting-related notifications shall be generated for events such as acceptance, cancellation, or rescheduling.

7. Use Case : Receive Notifications

I. Introduction : Receive Notifications use case allows students to receive notifications about important events related to their queries, document submissions, resolutions, and meetings. Notifications keep students informed about changes or actions that require their attention.

II. Actors :

1. **Primary Actors :** Students
2. **Supporting Actors :** TIC/Admin
3. **Supporting Services :** Notification Service

III. Pre - Conditions :

1. Student is authenticated.
2. The student has a query, document submission, or meeting associated with the system.
3. A notification-triggering event has occurred.
4. Notification service is available in the system.

IV. Post - Conditions :

1. The notification is generated and delivered to the student.
2. The notification is stored in the student's email.
3. The student can view the notification and any associated information.

V. Flow of Events :**Basic Flow :**

1. A relevant event occurs in the system.
2. System identifies the student associated with the event.
3. System generates an appropriate notification.
4. Notification is delivered through the email services.
5. The notification is displayed in the student's dashboard/notification/email section.
6. Student opens the notification.

Alternate Flow :**A1. Email delivery failure**

1. System attempts to send an email notification.
2. Email delivery fails.
3. System records the failure.
4. The notification remains available through the student's dashboard.

VI. Special Requirements

1. Notifications shall only be accessible to the intended recipient.
2. Important events such as query resolution and meeting changes should generate notifications promptly.
3. Notifications shall be stored so students can view previous notifications.
4. Notification delivery failures shall not result in loss of the underlying query, resolution, or meeting information.
5. Sensitive student information shall not be unnecessarily included in email notifications.
6. The system should support both in-app notifications and email notifications.

3.4 Admin Use Cases**1. Use Case : View New Queries**

I. Introduction : View New Queries use case allows an authenticated Admin to view newly submitted and unresolved student queries from a centralized dashboard.

II. Actors :

1. **Primary Actors :** Admin
2. **Supporting Actors :** Student

III. Pre-Conditions

1. Admin is authenticated.
2. Admin has the required permission to access query management.
3. At least one student query has been submitted, or the system is available to display the query list.

IV. Post - Conditions :

1. Admin can view the list of new queries.
2. Admin can view the details of a selected query.
3. The query remains available for further processing.
4. Viewing a query does not itself change its resolution.

V. Flow of Events :**Basic Flow**

1. Admin logs into the system.
2. Admin opens the Query Management section.
3. System retrieves newly submitted and relevant unresolved queries.
4. System displays the queries in the Administrator dashboard.
5. Administrator can view summary information such as: Student Name and relevant academic details, query description, submission date and time and current status.

6. Administrator selects a query.
7. System displays the complete query details.
8. Administrator reviews the query and determines the appropriate next action.

Alternate Flow

A1. No new queries

1. Administrator opens the Query Management section.
2. System finds no new queries.
3. System displays an appropriate message indicating that there are currently no new queries.

A2. Query information cannot be retrieved

1. System encounters an error while retrieving the query list or details.
2. System displays an appropriate error message.
3. Administrator may retry the request.

A3. Query has already been forwarded

1. Administrator selects a query that has already been forwarded to a TIC or another authorized handler.
2. System displays the assignment information.
3. Administrator can view the query but cannot perform actions restricted to the assigned authority.

Special Requirements

1. Only authorized Admins will be able to access the new-query dashboard.
2. Admins shall only access student information permitted by their role.
3. Queries shall be displayed with clear status and assignment information.
4. Sensitive student information and documents shall be protected from unauthorized access.
5. The dashboard should update sufficiently quickly to reflect newly submitted queries.

2. Use Case : Monitor and Resolve Queries directly

I. Introduction : Monitor and Resolve Queries directly use case allows an authenticated Admin to monitor a student query, review its details, and provide a resolution directly when the query falls within the Admin's authority. The resolution is recorded and communicated to the student and may be also updated into the institutional knowledge base.

II. Actors :

1. **Primary Actors :** Admin
2. **Supporting Actors :** Students
3. **Supporting Services :** Notification Service, Knowledge Base

III. Pre - Conditions :

1. Admin is authenticated and authorized to manage queries.
2. The query has been submitted and is available to the Admin.
3. The query has not already been resolved or closed.
4. Admin has sufficient information and authority to resolve the query directly.

IV. Post - Conditions :

5. The query is marked as **Resolved**.

1. The resolution is stored and associated with the query.
2. The student can view the resolution through their dashboard.
3. The student is notified of the resolution.
4. Where applicable, the verified resolution can be added or used to update the institutional knowledge base.

V. Flow of Events :

Basic Flow

1. Admin opens the Query Management dashboard.
2. Admin selects a query from the list of new or unresolved queries.
3. System displays the complete query details and relevant student information.
4. Admin reviews the query and determines that it can be resolved directly.
5. Admin enters the appropriate resolution or answer.
6. System validates the resolution information.
7. Admin submits the resolution.
8. System records the resolution along with the resolving authority and timestamp.
9. System changes the query status to **Resolved**.
10. System makes the resolution available to the student.
11. System sends a notification to the student.
12. Where appropriate, the verified resolution is made available for knowledge-base update.

Alternate Flow :

A1. Insufficient information or documents needed

1. Admin determines that additional information is required.
2. Admin may request clarification or supporting documents from the student.
3. System records the request.
4. Student is notified.
5. Query remains open until the required information is received.

A2. Query requires TIC's assistance

1. Admin determines that the query is department-specific or outside their authority.
2. Admin does not resolve the query directly.
3. Admin forwards the query to the appropriate TIC.
4. The query follows the **Use Case 3 - Forward Query to TIC** workflow.

VI. Special Requirements

1. Only authorized Admins shall be permitted to resolve administrative queries.
2. A resolution shall be associated with the corresponding query and resolving authority.
3. Student information displayed during resolution shall follow role-based access restrictions.
4. Resolutions containing verified institutional information should be eligible for knowledge-base updates.
5. Knowledge-base updates shall not occur from an unverified or draft response.
6. The student shall receive a notification after successful resolution.

3. Use Case : Forward Query to TIC

I. Introduction : Forward Query to TIC use case allows an authenticated Admin to forward a student query to the appropriate TIC when the query is department-specific or cannot be resolved directly by the Admin.

II. Actors :

1. **Primary Actors :** Admin
2. **Supporting Actors :** TIC
3. **Supporting Services :** Notification Services

III. Pre-Conditions :

1. Admin is authenticated and authorized to manage queries.
2. A student query is available for review.
3. The query has not already been resolved or closed.
4. The relevant departmental TIC is available in the system.

IV. Post-Conditions :

1. The query is assigned to the appropriate TIC.
2. The query status is updated to **Forwarded to TIC**.
3. The TIC can view and process the assigned query.
4. The student can see the updated query status.
5. Appropriate notifications are generated.

V. Flow of Events :

Basic Flow

1. Admin opens the Query Management dashboard.
2. Admin selects a query requiring departmental TICs assistance.
3. System displays the query details and relevant student information.
4. Administrator identifies the appropriate TIC.
5. System displays the forwarding details for confirmation.
6. Admin confirms the assignment.
7. System assigns the query to the selected TIC.
8. System updates the query status to **Forwarded to TIC**.
9. The query becomes available in the TIC's dashboard.
10. System notifies the TIC about the new assigned query.
11. System notifies the student that the query has been forwarded to the relevant TIC.

Alternate Flow

A1. Incorrect or unavailable TIC

1. Admin attempts to forward the query to a TIC who is unavailable or inactive.
2. System prevents the assignment.
3. System displays the available authorized TICs or authority for the relevant department.
4. Admin selects an appropriate authority.

A2. Assignment failure

1. System fails to save the assignment.
2. System informs the Admin that the query could not be forwarded.
3. Query remains unassigned or retains its previous assignment.
4. Admin may retry the operation

A3. Admin cancels forwarding

1. Admin selects a TIC but does not confirm the assignment.
2. Admin cancels the operation.
3. Query remains in its previous status and assignment.

Special Requirements :

1. Only authorized Admin shall be permitted to forward queries.
2. A query shall only be forwarded to an authorized and active TIC.
3. The system (AI/RAG engine) should suggest TIC selection based on the student's department and query category where applicable.
4. The system shall prevent unauthorized users from accessing forwarded queries.
5. The TIC shall receive a notification when a new query is assigned.
A query could be simultaneously assigned to multiple TICs as per requirement of the query.

4. Use Case : Manage and Maintain Knowledge Base

I. Introduction : Manage and Update/Deactivate Knowledge Base use case allows an authenticated Administrator to manage the institutional knowledge base by adding new verified information, updating existing information, and deactivating outdated or incorrect information. The knowledge base provides the verified source used by the AI/RAG system to answer student queries.

II. Actors :

1. **Primary Actors :** Admin
2. **Supporting Services :** AI/RAG Engine, Knowledge Base

III. Pre- Conditions :

1. Admin is authenticated and authorized to manage the knowledge base.
2. The knowledge base is available.
3. Information being added or updated has been verified by an authorized institutional authority/admin and validated by the system.

IV. Post- Conditions :

1. New verified information is added to the knowledge base or
2. Existing information is updated with the latest verified information or
3. Outdated/incorrect information is deactivated.
4. The AI/RAG system can use the updated active information for future queries.

V. Flow of Events :**Basic Flow**

1. Admin opens the Knowledge Base Management/database section.
2. Admin selects an operation : Add /Update/ Deactivate Information
3. System displays the appropriate confirmation interface.
4. Administrator enters or modifies the required information.
5. System validates the information.
6. Admin confirms the operations.
7. System stores the change.
8. System updates the active knowledge base used by the AI/RAG system.

9. System confirms that the operation was completed successfully.

Special Requirements :

1. Only authorized Admin users shall be able to add, modify, deactivate, or archive Knowledge Base information.
2. Active knowledge-base entries shall contain verified institutional information.
3. The system shall distinguish between active, inactive, and outdated information.
4. Archived, inactive, or expired information shall not be used for normal AI/RAG responses.
Updates shall not require retraining the underlying LLM and the updated knowledge should become available through the RAG retrieval layer.

3.5 TIC Use Cases**1. Use Case : View Assigned Queries**

I. Introduction : View Assigned Queries use case allows an authenticated TIC to view and review student queries that have been assigned to their department.

II. Actors :

1. **Primary Actors :** TIC
2. **Supporting Actors :** Student, Admin

III. Pre-Conditions :

1. TIC is authenticated.
2. TIC has the required permission to access departmental queries.
3. At least one query has been assigned or forwarded to the TIC, or the system is available to display the query list.

III. Post-Conditions :

1. TIC can view the list of queries assigned to them.
2. TIC can view the details of a selected query.
3. The query remains available for further processing.
4. Viewing the query does not itself resolve or close it.

IV. Flow of Events :**Basic Flow**

1. TIC logs into the system.
2. TIC opens the **Assigned Queries** section.
3. System retrieves queries assigned to the TIC.
4. System displays the assigned queries along with relevant information, such as Student name, Department, Academic Year, Query Description, Submission date, Current Status, Assignment Date.
5. TIC selects a query.
6. System displays the complete query details.
7. TIC reviews the query and determines the appropriate next action.

Alternate Flow**A1. No assigned queries**

1. TIC opens the Assigned Queries section.

2. System finds no queries assigned to the TIC.
3. System displays an appropriate message indicating that there are currently no assigned queries.

A2. Query has already been resolved

1. TIC selects a query that has already been resolved.
2. System displays the query and its resolution.
3. TIC can view the information but cannot modify it unless authorized.

A3. Query information cannot be retrieved

1. System encounters an error while retrieving the assigned queries.
2. System displays an appropriate error message.
3. TIC may retry the request.

Special Requirements

1. Only authenticated and authorized TICs shall be able to access assigned queries.
2. A TIC shall only be able to view queries assigned to them or their department.
3. Student information shall be displayed according to the TIC's access permissions.
4. Sensitive documents shall only be accessible to authorized users.

2. Use Case : Review and Resolve departmental queries

I. Introduction : Review and Resolve departmental queries use case allows an authenticated TIC to review and resolve student queries assigned to their department.

II. Actors :

1. **Primary Actors :** TIC
2. **Supporting Actors :** Student, Admin
3. **Supporting Services :** Notification Service, Knowledge Base

III. Pre-Conditions :

1. TIC is authenticated.
2. The query has been assigned to the TIC or their department.
3. The query has not already been resolved.
4. The TIC has sufficient authority to handle the query.

IV. Post-Conditions :

1. The query is resolved and marked **Resolved** or
2. Additional documents are requested from the student or
3. A meeting is scheduled when direct interaction is required.
4. The student is notified of the relevant update.
5. A verified departmental resolution may be considered for addition or update in the knowledge base.

V. Flow of Events :**Basic Flow**

1. TIC opens the **Assigned Queries** section.
2. TIC selects a departmental query.
3. System displays the query details and relevant student information.

4. TIC reviews the query and determines the appropriate course of action.
5. TIC enters a verified departmental resolution.
6. TIC submits the resolution.
7. System validates and records the resolution.
8. System associates the resolution with the query.
9. System changes the query status to **Resolved**.
10. System makes the resolution available to the student.
11. System notifies the student about the resolution.
12. Where applicable, the verified resolution is made available for knowledge-base updating.

Alternate Flow

A1. Additional documents are required

1. TIC determines that documents are required to process the query.
2. TIC selects **Need Document**.
3. TIC specifies the required document and any relevant instructions.
4. System records the document request.
5. Query status is changed to **Documents Required**.
6. Student is notified.
7. Student proceeds to **Student Use Case 5 - Upload Requested Documents**.
8. Teacher proceeds to **Teacher Use Case 3 and 4**.

A2. Submitted documents are insufficient

1. TIC reviews documents submitted by the student.
2. TIC determines that the documents are incomplete, invalid, or insufficient.
3. TIC requests replacement or additional documents.
4. System notifies the student.
5. The query remains open until the required documents are provided.

Special Requirements

1. Only authorized TICs shall be able to resolve queries assigned to their department.
2. TICs shall only access student information and documents relevant to their authorized queries.
3. Resolutions shall contain verified departmental information.
4. Uploaded documents shall be securely stored and accessible only to authorized personnel.
5. A verified departmental resolution shall not automatically become active knowledge-base information without the required validation process by the admin.

3. Use Case : Request Student Documents

I. Introduction : Request Student Documents use case allows an authenticated TIC to request documents from a student when additional documentation is required to process or resolve a departmental query. The TIC specifies the required documents and provides relevant instructions.

II. Actors :

1. **Primary Actors :** TIC
2. **Supporting Actors :** Student
3. **Supporting Services :** Notification Services, Document Storage Service.

III. Pre-Conditions :

1. TIC is authenticated and authorized to manage the relevant departmental query.
2. The query is assigned to the TIC.
3. The TIC has determined that one or more documents are required.
4. The query has not been resolved.

IV. Post-Conditions :

1. A document request is associated with the relevant query.
2. Query status is updated to **Documents Required**.
3. The student is notified of the document request.
4. The student can access the document-upload interface for the relevant query.

V. Flow of Events :**Basic Flow**

1. TIC opens an assigned departmental query.
2. TIC reviews the query and determines that additional documents are required.
3. TIC selects **Need Document**.
4. System displays the document-request form.
5. TIC specifies the required documents.
6. TIC provides any relevant instructions or remarks.
7. TIC submits the document request.
8. System validates the request.
9. System records the document request and associates it with the query.
10. System changes the query status to **Documents Required**.
11. System notifies the student about the requested documents.
12. Student can access the relevant query and proceed to **Student Use Case 5 - Upload Requested Documents**.

Alternate Flow**A1. Multiple documents required**

1. TIC identifies multiple required documents.
2. TIC adds each required document to the request.
3. TIC specifies which documents are mandatory, where applicable.
4. System records the complete document request.

A2. TIC cancels the request

1. TIC begins creating a document request.
2. TIC cancels the operation before submission.
3. No document request is created.
4. The query remains in its previous state.

A3. Invalid or incomplete request

1. TIC submits a request without specifying the required documents or other mandatory information.
2. System identifies the missing information.
3. System displays an appropriate validation message.
4. TIC completes the request and resubmits it.

Special Requirements :

1. Only authorized TICs shall be able to request documents for queries assigned to them.
2. The document request shall be associated with the specific student and query.
3. The TIC shall be able to specify the required document clearly.
4. The system should support marking documents as **Required**, where applicable.
5. Students shall only be able to view document requests associated with their own queries.
6. The system shall not expose documents or student information to unauthorized users.
7. The student shall receive a notification when documents are requested.

4. Use Case : Review Submitted Documents

I. Introduction : Review Submitted Documents use case allows an authenticated TIC to review documents submitted by a student in response to a document request.

II. Actors :

1. **Primary Actor:** TIC
2. **Supporting Actor:** Student
3. **Supporting Service:** Document Storage Service, Notification Service

III. Pre-Conditions :

1. TIC is authenticated and authorized to manage the relevant query.
2. A document request has been issued for the query through **TIC Use Case 3 Request Student Documents**.
3. The student has submitted one or more requested documents through **Student Use Case 5 Upload Requested Documents**.
4. The submitted documents are securely stored and available for review.

IV. Post-Conditions

1. The TIC has reviewed the submitted documents.
2. Each document is marked with an appropriate status, such as Accepted or Rejected.
3. The student is notified of the review outcome.
4. If all required documents are accepted, the TIC can continue processing the associated query.
5. If documents are insufficient, the student can be asked to submit additional or corrected documents.

V. Flow of Events**Basic Flow**

1. TIC opens the Assigned Queries section.
2. TIC selects a query with submitted documents.
3. System displays the document request and submitted documents.
4. TIC opens and reviews each submitted document.
5. TIC verifies whether the documents satisfy the requirements.
6. TIC records the review decision for each document.
7. System validates and stores the review decision.
8. System updates the document status accordingly.
9. If all required documents are accepted, the query status is updated to Documents Verified or Under Review

10. System records the TIC and review timestamp.
11. System notifies the student of the review outcome.
12. TIC continues processing the query.

Alternate Flow

A1. Document cannot be opened or retrieved

1. TIC attempts to access a submitted document.
2. System encounters an error or the file is unavailable.
3. System displays an appropriate error message.
4. TIC may retry the operation or request the student to resubmit the document.

A2. Document is incomplete

1. TIC identifies that a submitted document is incomplete, invalid, or does not satisfy the requirement.
2. TIC marks the document as Rejected.
3. TIC provides a reason or instruction for correction.
4. System records the decision.
5. Student is notified.
6. Student submits a corrected document through **Student Use Case 5 Upload Requested Documents**.
7. TIC reviews the replacement document.

A3. Additional document required

1. TIC determines that an additional document is necessary.
2. TIC creates an additional document request through **TIC Use Case 3 Request Student Documents**.
3. System updates the student's document requirements.
4. Student is notified.
5. Student submits the additional document.

Special Requirements :

1. Only authorized TICs shall be able to review documents associated with their assigned queries.
2. Documents shall be securely stored and transmitted.
3. The system shall prevent unauthorized users from accessing student documents.
4. The TIC shall be able to provide a reason when rejecting or requesting correction of a document.
5. The system should support common document formats and enforce configured file-size/type restrictions.
6. Students shall be notified when their documents are accepted, rejected, or when additional documents are required.

5. Use Case : View timetable

I. Introduction : View Timetable use case allows an authenticated TIC to view the timetable relevant to their department, including scheduled classes, available periods, and other relevant time slots.

II. Actors:

1. **Primary Actors :** TIC
2. **Supporting Services :** Timetable Service

III. Pre-Conditions :

1. TIC is authenticated.
2. Timetable information is available in the system or through the connected timetable service.

IV. Post-Conditions :

1. The requested timetable is displayed to the TIC.
2. The TIC can identify occupied and available time slots.
3. No timetable information is modified by this use case.

V. Flow of Events :**Basic Flow**

1. TIC opens the Timetable section from the dashboard.
2. System identifies the TIC's department and relevant academic schedule.
3. System retrieves the applicable timetable.
4. System displays the timetable in a structured format.
5. TIC can use the available information to proceed to the meeting-scheduling workflow.

Alternate Flow**A1. Timetable is unavailable**

1. System attempts to retrieve the timetable.
2. Timetable data is unavailable or the external service cannot be reached.
3. System displays an appropriate error message.
4. TIC can retry the request.

A2. No timetable is available for the selected period

1. TIC selects a date or academic period for which no timetable exists.
2. System informs the TIC that no timetable information is available.
3. TIC can select another period.

System Requirements

1. Only authorized TICs shall be able to access timetable information relevant to their role.
2. The timetable should be displayed in a clear day/week-based format.
3. Viewing the timetable shall not modify timetable records.

6. Use Case : Schedule/Offer/Cancel meetings

I. Introduction : Schedule/Offer/Cancel meetings use case allows an authenticated TIC to manage meetings with students by selecting available time slots, offering or scheduling meetings, and cancelling previously scheduled meetings when necessary.

II. Actors :

1. **Primary Actors :** TIC
2. **Supporting Actors :** Students
3. **Supporting Services :** Timetable Services, Notification Service

III. Pre-Conditions :

1. TIC is authenticated.
2. The student has an active query or departmental matter requiring a meeting.
3. Relevant timetable and availability information is accessible through **TIC Use Case 5 - View Timetable**.
4. The TIC has permission to schedule meetings with the student.

IV. Post-Conditions :

1. A meeting is successfully scheduled, cancelled, or otherwise updated.
2. The meeting details and status are stored in the system.
3. The student is notified of the meeting action.
4. Cancelled slots become available again, subject to system rules.

V. Flow of Events :**Basic Flow**

1. TIC opens the meeting section for a student's query.
2. System displays the student's relevant query information.
3. TIC views available time slots based on the timetable and existing bookings.
4. TIC selects a suitable date and time.
5. System verifies that the selected slot is still available.
6. TIC enters or confirms meeting details such as date, time, duration, and meeting mode/location.
7. TIC selects **Schedule Meeting**.
8. System creates the meeting record.
9. System assigns the appropriate meeting status, such as **Booked**.
10. The selected slot is reserved where applicable.
11. System notifies the student.
12. Student can view the meeting details through the dashboard.

Alternate Flow**A1. Timetable Service is unavailable**

1. System attempts to verify the selected slot.
2. Timetable Service is unavailable.
3. System prevents confirmation if availability cannot be reliably verified.
4. TIC is informed and can retry later.

A2. TIC cancels a meeting

1. TIC selects an existing proposed or booked meeting.
2. TIC chooses **Cancel Meeting**.
3. TIC provides a cancellation reason, where required.
4. System changes the meeting status to **Cancelled**.
5. The reserved slot is released.
6. Student is notified of the cancellation